WORK IN PROGRESS

# A Self-Determination Theory-based Career Counseling Chatbot: Motivational Interactions to Address Career Decision-Making Difficulties and Enhance Engagement

**HYERIM HAN**, Seoul National University of Science and Technology, Seoul, South Korea

**BOGYEOM PARK**, Seoul National University of Science and Technology, Seoul, South Korea

**KYOUNGWON SEO**, Seoul National University of Science and Technology, Seoul, South Korea

**Open Access Support** provided by:

**Seoul National University of Science and Technology**



**Published:** 25 April 2025

**Citation in BibTeX format**

CHI EA '25: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems
*April 26 - May 1, 2025*
*Yokohama, Japan*

**Conference Sponsors:**
SIGCHI


https://doi.org/10.1145/3706599.3720286
ISBN: 9798400713958

# A Self-Determination Theory-based Career Counseling Chatbot: Motivational Interactions to Address Career Decision-Making Difficulties and Enhance Engagement


Hyerim Han
Seoul National University of Science and Technology
Seoul, Republic of Korea
hanhyerim@seoultech.ac.kr

Bogyeom Park
Seoul National University of Science and Technology
Seoul, Republic of Korea
bogyeom@seoultech.ac.kr

Kyoungwon Seo
Seoul National University of Science and Technology
Seoul, Republic of Korea
kwseo@seoultech.ac.kr



## Abstract

Post-college unemployment represents a significant social problem, driven by graduates' career decision-making difficulties. Many individuals seek career counseling, but most methods focus on information delivery rather than motivation. Without sufficient motivation, students often struggle to make informed decisions. We aim to demonstrate the effectiveness of a self-determination theory-based career counseling chatbot (SDT chatbot) by developing and comparing two chatbots: SDT chatbot and a general career counseling chatbot (non-SDT chatbot). An experiment with 20 university students revealed that the SDT chatbot was more effective in addressing career decision-making difficulties. Semi-structured interviews showed that integrating competence, relatedness, and autonomy into the chatbot design facilitated the resolution of career decision-making difficulties. However, no significant differences were observed in engagement between the two chatbots. The analysis identified four aspects of chatbot interaction (crystallization, self-reflection, revalidation, and epistemic curiosity) that promote engagement. Based on these findings, we propose an SDT chatbot design for career counseling.



## CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**; • **Computing methodologies → Artificial intelligence**; • **Applied computing → Education**.

## Keywords

Self-determination theory, Large Language Models, Career counseling, Motivation, Career decision-making

**ACM Reference Format:**
Hyerim Han, Bogyeom Park, and Kyoungwon Seo. 2025. A Self-Determination Theory-based Career Counseling Chatbot: Motivational Interactions to Address Career Decision-Making Difficulties and Enhance Engagement. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3706599.3720286




ACM ISBN 979-8-4007-1395-8/25/04
https://doi.org/10.1145/3706599.3720286


## 1 Introduction

Career decision-making is a crucial process that shapes both personal and professional trajectories, directly influencing life satisfaction, psychological stability, and economic success [7, 19]. To make informed career decisions, individuals must identify and address the difficulties and underlying factors that hinder their decision-making process [18, 28]. Career counseling has long been recognized as one of the most effective methods for overcoming career decision-making difficulties and facilitating self-awareness and informed decision-making among students [30, 40]. In this counseling process, theoretical frameworks tailored to guide decision-making, such as Harren's model of career decision-making for college students, are widely utilized [21]. While this model supports students in making step-by-step decisions, it primarily focuses on providing information and fostering rational decision-making, which limits its ability to enhance students' intrinsic motivation [20]. Moreover, due to constraints in counseling personnel and resources, providing continuous, real-time, and personalized feedback remains a challenge [31, 41]. Consequently, research on advanced technologies that can support and enhance career counseling is essential for addressing these limitations effectively.

To overcome these challenges, recent advancements in large language model (LLM)-based chatbots have introduced new possibilities for career counseling [1, 15]. LLM-based chatbots can address many of the resource limitations associated with traditional counseling by offering personalized guidance and tailored career planning support, thereby improving the accessibility and scalability of career counseling services [13]. However, while these chatbots excel in providing information, they often fail to foster intrinsic motivation—an essential factor in effective career counseling—by reducing career decision-making difficulties and encouraging sustained engagement [8, 37, 43]. Without adequate intrinsic motivation, students may struggle to explore their options thoroughly and make well-informed career decisions [24].

A promising solution to bridge this gap is the application of self-determination theory (SDT), a well-established framework for understanding and enhancing motivation. SDT emphasizes the fulfillment of three fundamental psychological needs—competence, relatedness, and autonomy—as key drivers of intrinsic motivation and sustained engagement [35]. The principles of SDT align well with the objectives of career counseling by fostering self-efficacy and empowering students to make independent, informed decisions [2]. Despite its potential, the application of SDT in career counseling remains largely unexplored [22]. Therefore, empirical research is needed to examine the impact of chatbots incorporating SDT principles on the career decision-making process. This study aims to evaluate the effectiveness of an SDT-based career counseling chatbot (SDT chatbot) in comparison to a traditional information-providing chatbot (non-SDT chatbot). Specifically, it investigates whether the SDT chatbot alleviates career decision-making difficulties while enhancing students' engagement and motivation in the career decision-making process.

Both chatbots are designed based on Harren's model of career decision-making for college students; however, while the non-SDT chatbot follows conventional career counseling methods by focusing primarily on providing information and career recommendations, the SDT chatbot is designed to fulfill fundamental psychological needs—competence, relatedness, and autonomy—through carefully crafted prompts.

To assess the effectiveness of the SDT chatbot, this study employs a mixed-methods research design, combining experimental analysis with semi-structured interviews. By doing so, it seeks to verify the chatbot's impact on career decision-making and gain deeper insights into students' subjective experiences. This study addresses the following research questions:

- RQ1: What impact does an SDT chatbot have on reducing career decision-making difficulties?
- RQ2: How does an SDT chatbot influence student engagement and motivation during the career decision-making process?

## 2 Related Work

### 2.1 LLM-based career counseling

Traditional career counseling plays a crucial role in helping students explore their aptitudes, understand career options, and make informed decisions. However, several challenges limit the accessibility and effectiveness of traditional counseling services. Consequently, many researchers have recognized the potential of LLM in the context of career counseling [4]. Previous studies on career counseling chatbots have primarily focused on recommending suitable careers for students and providing career-related information [17, 23, 25]. However, most of these studies have not progressed toward designing and evaluating counseling models that systematically foster sustained engagement and intrinsic motivation among students [5, 16]. Motivation serves as a psychological mechanism that goes beyond mere information provision, encouraging students to actively engage in and immerse themselves in the career decision-making process [33]. Therefore, this study proposes an LLM-based career counseling chatbot that integrates chatbot elements aligned with a motivational theoretical framework.

### 2.2 Self-determination theory

SDT posits that both intrinsic motivation and well-internalized extrinsic motivation contribute to various positive educational outcomes, particularly when individuals' fundamental psychological needs are met [35]. According to SDT, three core psychological needs drive motivation: competence, which pertains to experiencing mastery and effectiveness; relatedness, which involves feeling meaningfully connected to others; and autonomy, which reflects the desire to act with volition. Effectively integrating SDT into various systems has been shown to enhance motivation and improve outcomes [14, 39]. As a result, there has been a growing effort to incorporate SDT principles into chatbot design [2, 42]. However, the potential of SDT remains largely underutilized in LLM-based career counseling. This study aims to explore how integrating SDT into LLM-based career counseling can enable chatbots to foster autonomy, competence, and relatedness while more effectively supporting students' self-determined motivation.

## 3 Materials and Method

### 3.1 Participants

A total of 20 participants (10 males and 10 females; average age of 25.00 ± 1.55 years) were recruited from the university community. Participants received a thorough explanation of the experimental procedures, and informed consent was obtained. A total of 20 participants were assigned to either SDT chatbot condition (n = 10; average age 25.00 ± 1.65 years) or non-SDT chatbot condition (n = 10; average age 26.00 ± 1.36 years). Each group included seven male and three female participants. Basic demographic characteristics, including gender and age, were compared between both chatbot conditions using a chi-square test and two-sided Mann-Whitney U test. The results indicated no significant difference in gender ($\chi^2(1) = 1, p = 1.00$) or age ($p = 0.12$) between both chatbot conditions. The study's procedures and participant recruitment were approved by the Institutional Review Board at Seoul National University of Science and Technology (IRB approval number: 2023-0021-01).

### 3.2 Two different chatbot conditions

This study examined how an SDT chatbot design affects students' career decision-making difficulties, engagement, and motivation. Both the SDT and the non-SDT chatbots follow the four-stage process of Harren's model, which is implemented using prompt engineering and built on the GPT-4o model. As shown in Table 1, Harren's career decision-making model provides a well-established framework that systematically supports career exploration and decision-making while serving as a foundation for incorporating SDT's motivational components [21]. Based on this framework, we designed an SDT chatbot to fulfill students' fundamental psychological needs across the three key components of SDT—competence, relatedness, and autonomy.

The interaction with the chatbots followed Harren's model, which consists of four stages: (1) awareness, (2) planning, (3) commitment, and (4) implementation. In the 'awareness' stage, both chatbots greet the user, prompting students to focus on their current situation. At the 'planning' stage, students engage with the chatbots to explore their options and, based on this, narrowing them down to one of the six alternatives recommended by chatbots. The SDT chatbot emphasizes the competence component here, as students need to feel confident in their ability to effectively navigate the decision-making process regarding their career paths. In the 'commitment' stage, students collaborate with the chatbot to plan a timeline leading up to graduation, preparing for their selected alternative. The SDT chatbot highlights the relatedness component here, as students need to receive feedback from others and make

**Table 1: Harren's model of career decision-making for college students**

| Stage | Explanation |
|---|---|
| **Awareness** | During this stage, the individual attends to the present self-in-situation and expands one's time perspective to include part of the past and the future in one's psychological present. |
| **Planning** | During this stage, there is an alternating, expanding and narrowing process of exploration and crystallization. The expanding aspect of exploration involves searching for information or data about the task and about the self-concept in relation to the task. |
| **Commitment** | During this stage, a future orientation involves developing plans for the implementation of commitments. These plans include specific action steps, contingency plans, and the awareness of a need for, and a search for, detailed information in order to implement the decision. |
| **Implementation** | During this stage, the individual is inducted into a new context, then reacts to the context, and finally is assimilated into the context. |

decisions through a process of trial and error with peers. Finally, in the 'implementation' stage, students take time to discuss and prepare for the challenges they may face as they transition into new careers. The SDT chatbot focuses on the autonomy component here, students need to stress the importance of students making their own choices while engaging in a new context. Each element of SDT is designed as follows as illustrated in Figure 1.

First, to foster competence, the SDT chatbot aimed to enhance self-efficacy, a crucial factor in developing a sense of capability [26]. During the Planning stage, the SDT chatbot considered the user's major and interests and career aspirations, explaining the strengths of each career option and why it would be a good fit for them. Additionally, it provided feedback to help users develop self-efficacy and confidence to make informed decisions [36]. On the other hand, the non-SDT chatbot provided only basic job descriptions and required qualifications, offering general information without connecting it to the user's strengths. Second, during the commitment stage, the SDT chatbot fosters a sense of relatedness by incorporating shared experiences and empathy with peers [3, 29]. It references seniors in the same department have gone through similar processes and procedures, helping users connect with the experiences of others. In contrast, non-SDT chatbot offered standard career advice without fostering connection. Lastly, during the implementation stage, the SDT chatbot encouraged users to make self-directed choices in their career decision-making process [12]. To support autonomy, the SDT chatbot avoided a directive tone and posed open-ended questions that enabled users to explore and make independent choices, emphasizing that the decision-making authority rested with them. In contrast, the non-SDT chatbot provided directive guidance with imposed standard goals and evaluations.

## 3.3 Career decision-making difficulties questionnaire

To assess the students' levels of career decision-making difficulties, this study employed the career decision-making difficulties questionnaire (CDDQ), which consists of 34 items. Developed by Gati et al. [18] based on decision theory and research on career indecision, the CDDQ assesses a range of specific difficulties individuals may face in making career decisions. Each item on the CDDQ represents a unique difficulty, with responses rated on a 9-point Likert scale ranging from 1 (Does not describe me at all) to 9 (Describes me very well), with higher scores indicating greater difficulty in career decision-making (see Appendix A for details).

## 3.4 Engagement and motivation questionnaire

To measure students' overall engagement and motivation during their interactions with chatbot, this study utilized the engagement and motivation questionnaire (EMQ), comprising 13 items [38]. The EMQ is structured as a hierarchical model, categorizing engagement and motivation into four main areas: behavioral engagement, emotional engagement, cognitive engagement, and motivation(see Appendix B, for details). Each item prompts participants to rate how accurately the statement reflects their experience on a 7-point Likert scale, from 1 ("Does not describe me at all") to 7 ("Describes me very well").

## 3.5 Experiment procedure

Participants were randomly assigned to one of two chatbot conditions, and the experiment proceeded in five phases [10]. (1) pre-CDDQ assessment for baseline levels, (2) career counseling with the assigned chatbot. All participants completed it, despite having the option to terminate at any time. (3) an EMQ assessment for engagement and motivation, (4) post-CDDQ assessment to evaluate changes in decision-making difficulties, and (5) semi-structured interviews to gain deeper qualitative insights into participants' experiences. This design captured immediate intervention effects, as career counseling impacts can emerge shortly after sessions [9]. The experiment lasted 33.6 ± 4.76 minutes, and took place in a quiet, private setting.

## 3.6 Data analysis

To compare significant differences between two chatbot conditions, we used two statistical tests—paired t-test, and two-sided Mann-Whitney U test—at a significance level of $\alpha = 0.05$. First, to evaluate the SDT chatbot's impact on career decision-making difficulties, pre-CDDQ and post-CDDQ scores were analyzed. Descriptive statistics were calculated, followed by a normality test, then a paired t-test for within-subjects comparisons. Second, to evaluate the SDT chatbot's impact on engagement and motivation, descriptive statistics were calculated for EMQ scores, and normality was tested.

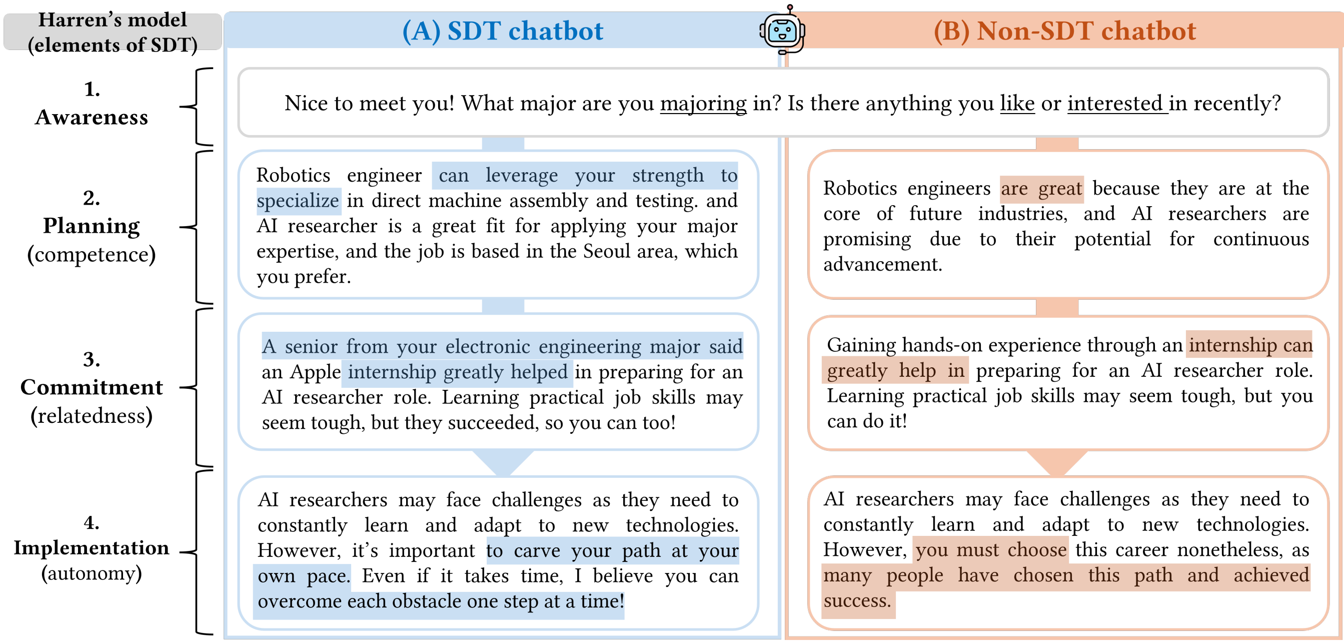


**Figure 1: Comparison of chatbot interactions with two different chatbot conditions (A) SDT chatbot : career counseling chatbot based on self-determination theory supports students' fundamental psychological needs for competence, relatedness and autonomy (B) non-SDT chatbot : general career counseling chatbot provided general career counseling without specific goal-setting, focusing solely on information delivery.**

However, since the data did not meet the assumption of normality, a two-sided Mann-Whitney U test was conducted to compare engagement and motivation. Lastly, motivation related to career decision-making difficulties and engagement was analyzed through a thematic analysis of participants' semi-structured interview responses [6, 11], based on the SDT framework (see Section 2.2). The thematic analysis began with the first author coding the responses by stage and two additional authors independently reviewed and refined the codes.

**Table 2: The results of the career decision-making difficulties questionnaire in both chatbot conditions, presented as Mean ± SD**

| | N | Pre-CDDQ | Post-CDDQ | $p$-value |
|---|---|---|---|---|
| SDT chatbot | 10 | 4.54 ± 0.78 | 4.14 ± 0.79 | .001*** |
| non-SDT chatbot | 10 | 4.24 ± 1.04 | 4.22 ± 1.05 | .86 |

*Note.* $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$

# 4 Results

## 4.1 Career decision-making difficulties questionnaire result

An independent samples t-test revealed no significant difference in pre-CDDQ scores between SDT chatbot ($M = 4.54, SD = 0.78$) and non-SDT chatbot ($M = 4.24, SD = 1.04$) conditions ($t(9) = 0.71, p = 0.49$). Paired t-tests were conducted for each group to compare the differences between pre- and post-CDDQ scores under the two chatbot conditions. SDT chatbot showed a statistically significant decrease in CDDQ scores, from 4.54 ± 0.78 to 4.14 ± 0.79 ($p = 0.00052$). In non-SDT chatbot condition, CDDQ scores decreased from 4.24 ± 1.04 to 4.22 ± 1.05, but this change was not statistically significant ($p = 0.86$). Therefore, while the SDT chatbot demonstrated a statistically significant reduction in CDDQ scores, no significant difference was observed in the non-SDT chatbot (see Table 2).

## 4.2 Engagement and motivation questionnaire result

Based on the results presented in Table 3, SDT chatbot (n = 10) and non-SDT chatbot (n = 10) were compared across four areas: behavioral engagement, emotional engagement, cognitive engagement, and motivation.

For behavioral engagement, SDT chatbot scored 6.17 ± 0.71, compared to 5.33 ± 1.37 for the non-SDT chatbot ($p = 0.21$). In terms of emotional engagement, SDT chatbot and non-SDT chatbot were 5.20 ± 1.53 and 4.73 ± 1.66, respectively ($p = 0.52$). Regarding cognitive engagement, SDT chatbot scored 5.70 ± 1.27, while non-SDT chatbot scored 5.37 ± 1.14 ($p = 0.27$). Pertaining to motivation, SDT chatbot scored 4.70 ± 0.81, compared to 4.43 ± 0.76 for non-SDT chatbot ($p = 0.29$). Overall, SDT chatbot showed slightly higher scores compared to non-SDT chatbot. However, the differences between two chatbot conditions were not statistically significant.

**Table 3: The results of the engagement and motivation questionnaire in both chatbot conditions, presented as Mean ± SD**

| | SDT Chatbot | non-SDT Chatbot | $p$-value |
|---|---|---|---|
| Behavioral engagement | 6.17 ± 0.71 | 5.33 ± 1.37 | .21 |
| Emotional engagement | 5.20 ± 1.53 | 4.73 ± 1.66 | .52 |
| Cognitive engagement | 5.70 ± 1.27 | 5.37 ± 1.14 | .27 |
| Motivation | 4.70 ± 0.81 | 4.43 ± 0.76 | .29 |

## 4.3 Motivational interaction to address career decision-making difficulties

We conducted semi-structured interviews to determine if the elements incorporated into the design of the SDT chatbot helped participants reduce their career decision-making difficulties (see Table 4).

The first aspect is competence. Most participants (six out of 10) who interacted with SDT chatbot acknowledged that explaining the strengths of each career option and why it would be a good fit for them was highly beneficial for making career decisions. In contrast, participants (three out of 10) who used non-SDT chatbot expressed that the chatbot would have been more helpful if it had provided recommendations and information aligned with their personal abilities and traits. Participants (nine out of 20) recognized that enhancing competence significantly contributed to reducing career decision-making difficulties.

The second aspect is relatedness. Among the participants (four out of 10) who interacted with SDT chatbot noted that knowing their seniors had followed a similar process enhanced their trust in career decision and facilitated their career decision-making. Conversely, participants (three out of 10) who interacted with non-SDT chatbot mentioned that having examples or methods from actual seniors in the same field would have been more helpful. Participants (six out of 20) perceived that strengthening relatedness contributes significantly to career decision-making difficulties.

Finally, an aspect of autonomy was examined. One participant (four out of ten) who used the SDT chatbot felt that open-ended questions and suggestions that encouraged self-reflection and exploration fostered a sense of self-direction, ultimately enhancing their internal control. Participant also felt that it positively influenced their career decision-making process. On the other hand, participants (four out of 10) who used non-SDT chatbot reported discomfort with the experience, citing limited options, a sense of predetermined answers, and feeling as though they were being directed by the chatbot. Participants (six out of 20) acknowledged that fostering autonomy plays a significant role in enhancing career decision-making difficulties.

## 4.4 Motivational interaction to enhance engagement

To analyze the design elements of chatbots that motivate students' engagement, we conducted a thematic analysis of the semi-structured interviews. We identify four main factors in motivational interaction, such as Table 5.

Table 5 summarizes the key themes emerging from participants' motivational interactions across chatbot conditions, identifying recurring patterns of positive (+) and negative (-) experiences. Crystallization and self-reflection were present in both chatbots, resulting in no significant differences in motivation. In contrast, revalidation and epistemic curiosity were prominent only in the SDT chatbot, significantly enhancing engagement.

*4.4.1 Crystallization.* Crystallization, or the process of translating vague thoughts about the future into concrete plans, occurred across both chatbot conditions and did not significantly differentiate their impacts. For example, one participant (SDT #1) noted that when asked about their field of interest, they could clarify previously vague ideas. Similarly, another participant (SDT #4), who had concerns about their career path, found that the chatbot helped them narrow their interests and visualize their future, increasing motivation. Conversely, Participant in the non-SDT chatbot condition (non-SDT #1) said the chatbot helped clarify abstract ideas, while another (non-SDT #10) found consolidating their thoughts useful, though it did not drive further motivation.

*4.4.2 Self-reflection.* Self-reflection, which involves organizing future plans by revisiting the past, was another factor observed in both chatbot conditions, with no substantial difference in its motivational impact. For instance, one participant (SDT #7) noted that the chatbot helped them narrow future options to two distinct fields, prompting self-reflection and clearer thinking. Similarly, another participant (SDT #10) explained that revisiting and structuring past ideas with the chatbot increased their motivation to work harder. In contrast, participant using the non-SDT chatbot (non-SDT #7) realized that among the three key requirements for engineers, they met only one and recognized the need to further develop their skills. Another (non-SDT #10) found the chatbot's question unexpectedly difficult but appreciated the opportunity for independent thinking. In contrast, Revalidation and Epistemic curiosity revealed clear differences between the two chatbot conditions, significantly influencing engagement in the SDT chatbot group.

*4.4.3 Revalidation.* Revalidation, or reaffirming existing knowledge through a chatbot, played a key role in motivational interaction across chatbot conditions. Participants in the SDT chatbot study reported that even when the chatbot reiterated familiar information, it helped them reorganize their thoughts, boosting confidence and motivation. One participant (SDT #1) found the chatbot's structured presentation of familiar information helpful for taking action and setting direction. Another (SDT #10) said it clarified their existing knowledge, leading to the realization that "If I follow my plan, I can achieve my goals," which they found motivating. In contrast, participants using the non-SDT chatbot reported less influence because they repeat what they know. One (non-SDT #5) said it helped retrace thoughts but lacked new insights, while another (non-SDT #7) noted it merely reflected their existing plans, offering little additional support. In the end, both groups received the same information, but only the SDT chatbot turned revalidation into motivation for engagement.

**Table 4: Results of the chatbot design components under both chatbot conditions.**

| | SDT chatbot | non-SDT chatbot |
|---|---|---|
| **Competence** | Students perceive personalized recommendations aligned with individual characteristics and competencies to be highly valuable, as they are likely to contribute significantly to informed career decision-making in the future. | Students perceive incorporating personalized recommendations tailored to individual characteristics and competencies would likely provide greater support and contribute more effectively to future career decision-making. |
| **Relatedness** | Students perceive senior students' shared experiences as trustworthy, as these firsthand accounts provide reliable insights. This trust is beneficial in helping them make well-informed career decisions based on real-life examples and practical guidance. | Students perceive that concrete examples and specific methods, such as case studies of their seniors, are more helpful in guiding their decision-making process. They believe that seeing real-life experiences and step-by-step approaches taken by others who have navigated similar career paths provides practical insights and makes the decision-making process feel more tangible and achievable. |
| **Autonomy** | Students perceive that they can accomplish anything when the chatbot presents various suggestions, as they see it as a range of choices they can make on their own. This sense of autonomy is beneficial as it helps them feel capable of making career-decisions independently. | Students perceive that they are restricted in their choices when the chatbot emphasizes one suggestion over others, as it makes them feel as if the answer has already been predetermined. They dislike the sense of having a limited range of options guided by the chatbot, as it gives the impression that they are not truly making their own career decisions. |

**Table 5: Summary of the students' motivational interactions of large language model based career counseling chatbot**

| | SDT chatbot | non-SDT chatbot |
|---|---|---|
| **Crystallization** | (+) Students are motivated through conversations with chatbots to concretize vague thoughts about how to shape their future. Thoughts lead to motivation, which in turn leads to engagement. | (+) Students thought that conversations with chatbots could shape abstract ideas.<br>(-) However, thoughts did not lead to motivation. |
| **Self-reflection** | (+) Students were motivated because conversations with the chatbot encouraged self-reflection and helped them organize their thoughts effectively. | (+) Students recognized areas where they needed improvement and found it interesting that the chatbot suggested ways to address these gaps,<br>(-) but it focused solely on immediate actions and failed to lead to deeper self-reflection. |
| **Revalidation** | (+) Students are motivated by the increased confidence gained from recognizing numerous similarities to their own thoughts when they organize previously known information through conversations with chatbots. | (-) Students felt that conversations with the chatbot mainly reiterated information they already knew, which they did not find particularly helpful for their motivation. |
| **Epistemic Curiosity** | (+) Students are motivated by the chatbot's recommendation of unexpected options, prompting them to consider further exploration of these alternatives. | (-) Students perceive the role of the chatbot as being limited to providing unexpected information. |

*4.4.4 Epistemic curiosity.* Participants in the SDT chatbot study reported that the chatbot strengthened their motivation by helping them explore options they had never considered. For example, one participant (SDT #5) discovered a previously unconsidered option through the chatbot. This realization led them to see its potential in situations where they lacked practical experience, motivating them to try it out. Similarly, another participant (SDT #6) noted that an unexpected recommendation from the chatbot encouraged them to search for related information, demonstrating how unexpected insights sparked curiosity and motivation. In contrast, participants using the non-SDT chatbot perceived its role as limited to basic information delivery. One participant (non-SDT #5) felt that newly provided information was somewhat clichéd and lacked real value. Likewise, another participant (non-SDT #6) found the chatbot's responses predictable and unengaging, leading them to disregard the information altogether. In the end, both chatbots provided new information, but only participants who interacted with the SDT chatbot were motivated for engagement through epistemic curiosity.

# 5 Discussion and Conclusion

## 5.1 Implications and design recommendations

Our study aimed to identify career counseling chatbot design elements based on SDT that effectively address career decision-making difficulties and enhance engagement. First, CDDQ scores decreased for both chatbots post-experiment, but only the SDT chatbot showed a statistically significant reduction, while the non-SDT chatbot's decrease was insignificant. To understand this effect, we conducted semi-structured interviews. In the awareness stage, students using the SDT chatbot reported gaining relevant career information tailored to their competencies and learning how to integrate self-knowledge with career data. Competency-supporting design helped them better understand themselves and effectively synthesize career information, thereby alleviating decision-making difficulties. These findings align with prior research [44], which showed that LLM-driven interactions enhancing student competencies lead to improved outcomes. During the planning stage, students felt supported by their own choices, particularly through senior students' stories, despite fears of failure and uncertainty about next steps. Relatedness-supporting design fostered trust and confidence in their career choices, further easing decision-making challenges. These results are consistent with previous research [32], which highlights the critical role of social support in career exploration and decision-making. In the implementation stage, students reported that decision-making became easier as they realized they had control over their choices. Autonomy-supporting design empowered them to make independent decisions, ultimately resolving career decision difficulties. These findings align with prior research [34], which demonstrated that open-ended questions and suggestions enhance autonomy by allowing users to explore options and make independent decisions. In conclusion, chatbot designs incorporating the three SDT elements—competency, relatedness, and autonomy—significantly reduced CDDQ scores, demonstrating their effectiveness in addressing career decision-making difficulties.

Second, EMQ scores showed no statistically significant difference between the two chatbots. However, the SDT chatbot exhibited slightly higher scores in all areas. As a result, we conducted semi-structured interviews, which led to the identification of four key factors that significantly influence students' motivation to engage. Thematic analysis revealed that both chatbot groups facilitated interactions that helped users clarify future plans and engage in self-reflection, leading to no statistically significant differences in overall engagement. However, the SDT chatbot promoted interactions that provided user revalidation and stimulated epistemic curiosity, resulting in relatively higher engagement than the non-SDT chatbot. These findings align with previous research [27], which suggests that user responses vary based on chatbot design. This study highlights the potential of SDT chatbots to enhance motivation and engagement by fostering interactions that encourage revalidation and epistemic curiosity.


# Acknowledgments

This work was supported by Institute of Information & communications Technology Planning & Evaluation(IITP) grant funded by the Korea government(MSIT) (RS-2023-00262158).


# References


[1] Tariq Alqahtani, Hisham A Badreldin, Mohammed Alrashed, Abdulrahman I Alshaya, Sahar S Alghamdi, Khalid Bin Saleh, Shuroug A Alowais, Omar A Alshaya, Ishrat Rahman, Majed S Al Yami, et al. 2023. The emergent role of artificial intelligence, natural learning processing, and large language models in higher education and research. *Research in social and administrative pharmacy* 19, 8 (2023), 1236–1242.

[2] Nagaletchimee Annamalai, Mohd Elmagzoub Eltahir, Samer H Zyoud, Deepa Soundrarajan, Bilal Zakarneh, and Najeh Rajeh Al Salhi. 2023. Exploring English language learning via Chabot: A case study from a self determination theory perspective. *Computers and Education: Artificial Intelligence* 5 (2023), 100148.

[3] Roy F Baumeister and Mark R Leary. 2017. The need to belong: Desire for interpersonal attachments as a fundamental human motivation. *Interpersonal development* (2017), 57–89.

[4] Mohit Bhatnagar. 2024. Enhancing career guidance through intent mining with large language models. In *International Conference on Artificial Intelligence in Education Technology*. Springer, 17–33.

[5] Rajesh Bhayana. 2024. Chatbots and large language models in radiology: a practical primer for clinical and research applications. *Radiology* 310, 1 (2024), e232756.

[6] Virginia Braun and Victoria Clarke. 2013. Successful qualitative research: A practical guide for beginners. (2013).

[7] Luara Carvalho, Luciana Mourão, and Clarissa Freitas. 2023. Career counseling for college students: Assessment of an online and group intervention. *Journal of Vocational Behavior* 140 (2023), 103820.

[8] Seungeon Cha, Martin Loeser, and Kyoungwon Seo. 2024. The impact of AI-based course-recommender system on students' course-selection decision-making process. *Applied Sciences* 14, 9 (2024), 3672.

[9] Youjie Chen and René F Kizilcec. 2024. Showing authentic examples of academic and career trajectories to influence college students' career exploration. *Journal of Vocational Behavior* 153 (2024), 104026.

[10] Doo Sung Choi, Jongyoul Park, Martin Loeser, and Kyoungwon Seo. 2024. Improving counseling effectiveness with virtual counselors through nonverbal compassion involving eye contact, facial mimicry, and head-nodding. *Scientific Reports* 14, 1 (2024), 506.

[11] Victoria Clarke and Virginia Braun. 2017. Thematic analysis. *The journal of positive psychology* 12, 3 (2017), 297–298.

[12] Edward L Deci and Richard M Ryan. 1985. The general causality orientations scale: Self-determination in personality. *Journal of research in personality* 19, 2 (1985), 109–134.

[13] Anuja Deshpande, Aryan Dubey, Arya Dhavale, Ankita Navatre, Uma Gurav, and Amit Kumar Chanchal. 2024. Implementation of an nlp-driven chatbot and ml algorithms for career counseling. In *2024 International Conference on Inventive Computation Technologies (ICICT)*. IEEE, 853–859.

[14] Hitesh Dhiman, Gustavo Alberto Rovelo Ruiz, Raf Ramakers, Danny Leen, and Carsten Röcker. 2024. Designing instructions using self-determination theory to improve motivation and engagement for learning craft. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–16.

[15] Wantong Du, Zhiying Zhu, Xinhui Xu, Haoyuan Che, and Shi Chen. 2024. CareerSim: Gamification design leveraging LLMs for career development reflection. In *Extended abstracts of the CHI conference on Human Factors in Computing Systems*. 1–7.

[16] Godson D'Silva, Megh Jani, Vipul Jadhav, Amit Bhoir, and Prithvi Amin. 2020. Career counselling chatbot using cognitive science and artificial intelligence. In *Advanced Computing Technologies and Applications: Proceedings of 2nd International Conference on Advanced Computing Technologies and Applications—ICACTA 2020*. Springer, 1–9.

[17] Luke K Fryer, Kaori Nakao, and Andrew Thompson. 2019. Chatbot learning partners: Connecting learning experiences, interest and competence. *Computers in human Behavior* 93 (2019), 279–289.

[18] Itamar Gati, Mina Krausz, and Samuel H Osipow. 1996. A taxonomy of difficulties in career decision making. *Journal of counseling psychology* 43, 4 (1996), 510.

[19] Itamar Gati, Nimrod Levin, and Shiri Landman-Tal. 2019. Decision-making models and career guidance. *International handbook of career guidance* (2019), 115–145.

[20] Frédéric Guay, Caroline Senécal, Lysanne Gauthier, and Claude Fernet. 2003. Predicting career indecision: A self-determination theory perspective. *Journal of counseling psychology* 50, 2 (2003), 165.

[21] Vincent A Harren. 1979. A model of career decision making for college students. *Journal of vocational behavior* 14, 2 (1979), 119–133.

[22] Joshua Howard, Marylène Gagné, Alexandre JS Morin, and Anja Van den Broeck. 2016. Motivation profiles at work: A self-determination theory approach. *Journal of vocational behavior* 95 (2016), 74–89.

[23] Adán José-García, Alison Sneyd, Ana Melro, Anaïs Ollagnier, Georgina Tarling, Haiyang Zhang, Mark Stevenson, Richard Everson, and Rudy Arthur. 2023. C3-IoC: A career guidance system for assessing student skills using machine learning and network visualisation. *International Journal of Artificial Intelligence in Education* 33, 4 (2023), 1092–1119.

[24] Florian E Klonek, Elisabeth Wunderlich, Daniel Spurk, and Simone Kauffeld. 2016. Career counseling meets motivational interviewing: A sequential analysis of dynamic counselor–client interactions. *Journal of Vocational Behavior* 94 (2016), 28–38.

[25] Terri Lee, Tianjie Zhu, Sophia Liu, Linh Trac, Zizhen Huang, and Yu Chen. 2021. CASExplorer: A conversational academic and career advisor for college students. In *Proceedings of the Ninth International Symposium of Chinese CHI*. 112–116.

[26] Hadassah Littman-Ovadia, Vered Lazar-Butbul, and Benny A Benjamin. 2014. Strengths-based career counseling: Overview and initial evaluation. *Journal of Career Assessment* 22, 3 (2014), 403–419.

[27] Ewa Luger and Abigail Sellen. 2016. " Like Having a Really Bad PA" The Gulf between User Expectation and Experience of Conversational Agents. In *Proceedings of the 2016 CHI conference on human factors in computing systems*. 5286–5297.

[28] Wei-Cheng Mau. 2001. Assessing career decision-making difficulties: A cross-cultural study. *Journal of Career Assessment* 9, 4 (2001), 353–364.

[29] Shery Mead, David Hilton, and Laurie Curtis. 2001. Peer support: a theoretical perspective. *Psychiatric rehabilitation journal* 25, 2 (2001), 134.

[30] Francis Milot-Lapointe, Réginald Savard, and Yann Le Corff. 2018. Intervention components and working alliance as predictors of individual career counseling effect on career decision-making difficulties. *Journal of Vocational Behavior* 107 (2018), 15–24.

[31] Christine Mulhern. 2023. Beyond teachers: Estimating individual school counselors' effects on educational attainment. *American economic review* 113, 11 (2023), 2846–2893.

[32] Minh Pham, Bao Quoc Lam, and Anh Tuan Ngoc Bui. 2024. Career exploration and its influence on the relationship between self-efficacy and career choice: The moderating role of social support. *Heliyon* 10, 11 (2024).

[33] Gerry Rayner and Theo Papakonstantinou. 2020. The use of self-determination theory to investigate career aspiration, choice of major and academic achievement of tertiary science students. *International Journal of Science Education* 42, 10 (2020), 1635–1652.

[34] Johnmarshall Reeve and Hyungshim Jang. 2006. What teachers say and do to support students' autonomy during a learning activity. *Journal of educational psychology* 98, 1 (2006), 209.

[35] Richard M Ryan and Edward L Deci. 2020. Intrinsic and extrinsic motivation from a self-determination theory perspective: Definitions, theory, practices, and future directions. *Contemporary educational psychology* 61 (2020), 101860.

[36] Dale H Schunk. 1983. Ability versus effort attributional feedback: differential effects on self-efficacy and achievement. *Journal of educational psychology* 75, 6 (1983), 848.

[37] Kyoungwon Seo, Samuel Dodson, Negar M Harandi, Nathan Roberson, Sidney Fels, and Ido Roll. 2021. Active learning with online video: The impact of learning context on engagement. *Computers & Education* 165 (2021), 104132.

[38] Thitaree Tanprasert, Sidney S Fels, Luanne Sinnamon, and Dongwook Yoon. 2024. Debate chatbots to facilitate critical thinking on youtube: Social identity and conversational style make a difference. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–24.

[39] Tsai-Hsuan Tsai, Yung-Sheng Chang, Hsien-Tsung Chang, and Yu-Wen Lin. 2021. Running on a social exercise platform: Applying self-determination theory to increase motivation to participate in a sporting event. *Computers in Human Behavior* 114 (2021), 106523.

[40] Susan C Whiston, Thomas L Sexton, and David L Lasoff. 1998. Career-intervention outcome: A replication and extension of Oliver and Spokane (1988). *Journal of Counseling Psychology* 45, 2 (1998), 150.

[41] Chenoa S Woods and Thurston Domina. 2014. The school counselor caseload and the high school-to-college pipeline. *Teachers College Record* 116, 10 (2014), 1–30.

[42] Xi Yang and Marco Aurisicchio. 2021. Designing conversational agents: A self-determination theory approach. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–16.

[43] Aijia Yuan, Edlin Garcia Colato, Bernice Pescosolido, Hyunju Song, and Sagar Samtani. 2025. Improving workplace well-being in modern organizations: A review of large language model-based mental health chatbots. *ACM Transactions on Management Information Systems* 16, 1 (2025), 1–26.

[44] Lili Zhang and Weiyu Zhang. 2024. Integrating large language models into project-based learning based on self-determination theory. *Interactive Learning Environments* (2024), 1–13.


# A Career decision-making difficulties questionnaire

Please rate the statements regarding your difficulties in making career decisions from strongly disagree to strongly agree (9-point Likert Scale):

(1) I know that I have to choose a career, but I don't have the motivation to make the decision now (I don't feel like it)
(2) Work is not the most important thing in one's life and therefore the issue of choosing a career doesn't worry me much.
(3) I believe that I do not have to choose a career now because time will lead me to the right career choice.
(4) It is usually difficult for me to make decisions.
(5) I usually feel that I need confirmation and support for my decisions from a professional person or somebody else I trust.
(6) I am usually afraid of failure.
(7) I like to do things my own way.
(8) I expect that entering the career I choose will also solve my personal problems.
(9) I believe there is only one career that suits me.
(10) I expect that through the career I choose I will fulfill all my aspirations.
(11) I believe that a career choice is a one-time choice and a life-long commitment.
(12) I always do what I am told to do, even if it goes against my own will.
(13) I find it difficult to make a career decision because I do not know what steps I have to take.
(14) I find it difficult to make a career decision because I do not know what factors to take into consideration.
(15) I find it difficult to make a career decision because I don't know how to combine the information I have about myself with the information I have about the different careers.
(16) I find it difficult to make a career decision because I still do not know which occupations interest me.
(17) I find it difficult to make a career decision because I am not sure about my career preferences yet (for example, what kind of a relationship I want with people, which working environment I prefer).
(18) I find it difficult to make a career decision because I do not have enough information about my competencies (for example, numerical ability, verbal skills) and/or about my personality traits (for example, persistence, initiative, patience).
(19) I find it difficult to make a career decision because I do not know what my abilities and/or personality traits will be like in the future.
(20) I find it difficult to make a career decision because I do not have enough information about the variety of occupations or training programs that exist.
(21) I find it difficult to make a career decision because I do not have enough information about the characteristics of the occupations and/or training programs that interest me (for example, the market demand, typical income, possibilities of advancement, or a training program's perquisites).
(22) I find it difficult to make a career decision because I don't know what careers will look like in the future.

(23) I find it difficult to make a career decision because I do not know how to obtain additional information about myself (for example, about my abilities or my personality traits).
(24) I find it difficult to make a career decision because I do not know how to obtain accurate and updated information about the existing occupations and training programs, or about their characteristics.
(25) I find it difficult to make a career decision because I constantly change my career preferences (for example, sometimes I want to be self-employed and sometimes I want to be an employee).
(26) I find it difficult to make a career decision because I have contradictory data about my abilities and/or personality traits (for example, I believe I am patient with other people but others say I am impatient).
(27) I find it difficult to make a career decision because I have contradictory data about the existence or the characteristics of a particular occupation or training program.
(28) I find it difficult to make a career decision because I'm equally attracted by a number of careers and it is difficult for me to choose among them.
(29) I find it difficult to make a career decision because I do not like any of the occupations or training programs to which I can be admitted.
(30) I find it difficult to make a career decision because the occupation I am interested in involves a certain characteristic that bothers me (for example, I am interested in medicine, but I do not want to study for so many years).
(31) I find it difficult to make a career decision because my preferences can not be combined in one career, and I do not want to give any of them up (e.g., I'd like to work as a free-lancer, but I also wish to have a steady income).
(32) I find it difficult to make a career decision because my skills and abilities do not match those required by the occupation I am interested in.
(33) I find it difficult to make a career decision because people who are important to me (such as parents or friends) do not agree with the career options I am considering and/or the career characteristics I desire.
(34) I find it difficult to make a career decision because there are contradictions between the recommendations made by different people who are important to me about the career they recommend that I choose, or about what career characteristics should guide my decision.

## B Engagement and Motivation questionnaire

Rate the statements regarding your experience talking to the chatbot from strongly disagree to strongly agree (7-point Likert scale). In all statements, "the activity" refers conversation with chatbot:

### B.1 Behavior Engagement

(1) I concentrated during the conversation.
(2) I was persistent during the conversation.
(3) I want to find out more about my feelings and thoughts.

### B.2 Emotional Engagement

(1) I liked the conversation.
(2) When I shared my feelings and thoughts with the chatbot, I felt interested.
(3) The response I got from the chatbot was interesting.

### B.3 Cognitive Engagement

(1) During my conversation with the chatbot, I tried to explain my personal experience and worked towards a solution.
(2) I attempted to find new insights from the chatbot by mentally relating or contrasting them with my personal thoughts or experiences.
(3) During the conversation with the chatbot, I evaluated the relevance or usefulness of response provided.

### B.4 Motivation

(1) I do the activity because I think the activity is interesting. (Intrinsic motivation)
(2) I do the activity because I think this activity is good for me. (Identified regulation)
(3) I do the activity because I'm supposed to do it. (External regulation)
(4) There may be good reasons to do this activity, but personally I don't see any. (Amotivation)